

# Solution Requirements

|                      |                                 |
|----------------------|---------------------------------|
| <b>Date</b>          | 29 June 2026                    |
| <b>Team ID</b>       | XXXX                            |
| <b>Project Name</b>  | Credit Card Approval Prediction |
| <b>Maximum Marks</b> | 2 Marks                         |

## Solution Requirements

The system accepts applicant information such as personal details, employment status, income, and credit history. After validating the input data, the machine learning model analyzes the applicant's profile using historical credit card application data and predicts whether the application should be Approved or Rejected. The predicted result is then displayed through a simple user interface for the bank officer.

## Functional Requirements

| Requirement ID | Requirement     | Description                                                                               |
|----------------|-----------------|-------------------------------------------------------------------------------------------|
| FR-01          | User Input      | The system shall allow applicants to enter credit card application details.               |
| FR-02          | Data Validation | The system shall validate the entered information before processing.                      |
| FR-03          | Dataset Access  | The system shall retrieve historical applicant data from the dataset.                     |
| FR-04          | Prediction      | The machine learning model shall predict whether the application is approved or rejected. |
| FR-05          | Display Result  | The system shall display the prediction result to the bank officer.                       |

Non-Functional Requirements

| Requirement ID | Requirement | Description                                                               |
|----------------|-------------|---------------------------------------------------------------------------|
| NFR-01         | Performance | The system should generate predictions within a few seconds.              |
| NFR-02         | Accuracy    | The prediction model should provide high accuracy for approval decisions. |
| NFR-03         | Reliability | The system should consistently produce reliable results for valid inputs. |
| NFR-04         | Security    | Applicant information should be stored and processed securely.            |
| NFR-05         | Usability   | The application should provide a simple and user-friendly interface.      |

Software Requirements

| Software          | Purpose                       |
|-------------------|-------------------------------|
| Python            | Programming language          |
| Jupyter Notebook  | Model development and testing |
| VS Code / PyCharm | Code development              |
| Flask             | Web application framework     |
| Pandas            | Data manipulation             |
| NumPy             | Numerical computations        |
| Scikit-learn      | Machine learning algorithms   |

|            |                           |
|------------|---------------------------|
| Matplotlib | Data visualization        |
| Seaborn    | Statistical visualization |

Hardware Requirements

| Component        | Specification                                |
|------------------|----------------------------------------------|
| Processor        | Intel i3 or above                            |
| RAM              | Minimum 4 GB (8 GB Recommended)              |
| Storage          | Minimum 2 GB Free Space                      |
| Operating System | Windows / Linux / macOS                      |
| Internet         | Required for dataset download and deployment |